

BAYESIAN STATISTICS

STAT 6300

Fall 2026

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Portfolio 1: Your first Bayesian model

Launched: September 15

Due: September 25

The goal of this first assignment is to complete one iteration of the Bayesian workflow. You will begin with a question about a dataset, manipulate the dataset to make it suitable for analysis, write a probabilistic model (a “generative story”), then go through the standard loop: testing the implications of the model (prior-predictive), obtaining a posterior distribution (estimation), and making predictions to check the quality of the fit (posterior-predictive). This model should be deliberately simple, as the objective is to practice one repetition of the workflow and communicate your results clearly, without getting bogged down by technicalities.

Deliverables

- Report (PDF): hw1__lastname.pdf
- One portfolio slide (PDF): hw1__lastname_slide.pdf
- Code (zip): hw1__lastname_code.zip. Include reproducible code for your work. Stan is required for the model. Any language is fine for the rest of the code. Do not include data or results files.

Dataset

Choose one of the datasets provided with this brief, or bring your own dataset to this assignment. Each of the provided datasets is accompanied by a README detailing its provenance, variables, observational unit, and the data manipulation you are expected to do *before* modeling.

If you use your own data, give enough detail for a reader to understand how the observations came into existence. **Crucially:** You should be able to derive one numerical outcome for each observational unit, e.g., the test score of students, the measured concentration of a chemical at a certain location, and so on. This is the quantity we will model. Any other form of dataset is unsuitable for this first exercise.

Work

Report

The report should include the following headers. Each section can be quite short.

1. **Data construction and description.** Describe the dataset, its origin (citation if applicable), and its observational unit. Explain the work necessary to go from “raw” unprocessed data to numerical observations.

2. **Modeling question.** With the tools we have seen so far, we can model numerical observations as drawn from a distribution with a likelihood. Hence, we can ask questions like: What are the parameters of a distribution that best describe the data? Or: Do these parameters differ for different types of units? Think of a question of interest given the dataset, and write it down. Include a couple of sentences about why you chose this particular question and dataset.
3. **Model description.** Construct a generative model for the data. State the likelihood and prior in “model notation” and in words. Define every observed quantity, unknown parameter, and fixed quantity in the model, as well as their support. Explain why the chosen probability families are reasonable descriptions of the generative process. We have seen the Bernoulli, Binomial and Normal likelihoods in class. You may peruse [Stan’s function reference](#) to find other likelihoods if you judge these unsuitable for your data.
4. **Prior-predictive analysis.** Consider and simulate from *more than one* plausible prior choice before fitting the model. Simulate complete datasets with the same size and structure as the observed data. Show the simulated outcomes and, if applicable, identify values or datasets that you judge physically impossible, scientifically implausible, or otherwise inconsistent with the stated problem.

Explain anything you have changed to avoid those outcomes and why the final prior remains defensible. Finally, explain why this prior predictive distribution is a test of the model before seeing the outcomes and not already a fit to the observed data.
5. **Estimation.** Fit the model, with the final choice of prior. Show the posterior distribution of every unknown parameter. Report whether Stan produced warnings.
6. **Posterior predictive check.** Use draws from the posterior distribution to simulate replicated datasets. Compare the observed data with the replicated data using displays or summaries that reveal important features of the distribution. Explain the most important agreement or disagreement and what it says about the model.
7. **Interpretation.** Interpret the parameters in the units of the problem.
8. **Self-assessment** In two lines tell me what you’re least sure about and why and roughly how long this assignment took you to complete.

Portfolio slide

The slide must include

- a title
- one-sentence plain-language summary of the dataset
- model in sampling notation
- visual representation of the posterior distribution
- key scientific takeaway (interpretation)

Anything beyond that is up to you!

How feedback will be organized

Feedback will address the following questions.

1. Is the construction of the data transparent and reproducible?
2. Is the probability model coherent, with correct supports and justified likelihood and prior choices?
3. Does the prior predictive analysis test prior implications without being turned into a fit of the data?
4. Does the Stan code compile?
5. Do the report and personal slide communicate the conclusion clearly?